A Review of Augmented Observations in Hurricanes

BRIAN MCNOLDY

SENIOR RESEARCH ASSOCIATE

UNIVERSITY OF MIAMI
ROSENSTIEL SCHOOL of MARINE, ATMOSPHERIC & EARTH SCIENCE

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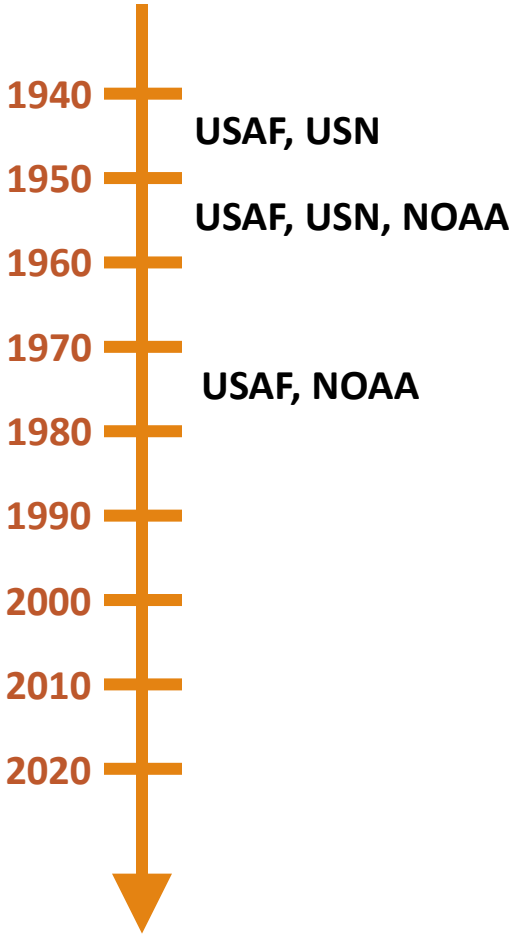
RENATO MOLINA, IVAN RUDIK, DIEGO CARDOSO

Hurricane Aircraft Reconnaissance

- This is the practice of flying in and around hurricanes with specially-instrumented aircraft. Data collected include flight-level wind, central pressure estimates, surface wind estimates, radar observations and derived wind fields, dropsonde observations, and so on.
- **For this project, "*augmented observations*" (AO) will refer specifically to dropsondes released from aircraft in and around hurricanes.**
  - A dropsonde is small expendable instrument package that once released from the aircraft, parachutes down to the ocean surface, collecting a vertical profile of temperature, moisture, wind, etc. along the way.

Atlantic Hurricane Reconnaissance History

- Mid 1940s (U.S. Air Force and Navy)
- Mid 1950s (U.S. Weather Bureau's National Hurricane Research Project)
- Mid 1960s (NHRP -> NHRL)
- Mid 1970s (U.S. Navy ceases hurricane reconnaissance)
- Mid 1980s (NHRL -> HRD)
- **Collecting data directly from the storm and its environment was immediately recognized to be beneficial to the human analyst/forecaster, as well as to computer models.**



Data Assimilation

- DA is a complex step in numerical weather prediction (NWP) in which observations are processed to create an optimal initial state for the forecast model.
- Observations can come from satellite, radar, ships, buoys, aircraft, radiosondes, dropsondes, surface stations.
- In principle, **a more accurate representation of the initial state of the atmosphere leads to a more accurate forecast by the model**.
- First used in the 1950s, as soon as NWP became operational.

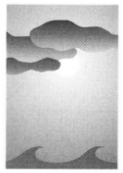
Data Denial Experiments

- For the past ~4 decades, “data denial” experiments have been conducted to evaluate the impact of a specified data type
  - Run the DA scheme twice: once with all available observations and once with a data type omitted.
  - Run the forecast model twice: use both sets of initial conditions.
- Not done routinely
  - Technical setup, computational expense, detailed comparison study

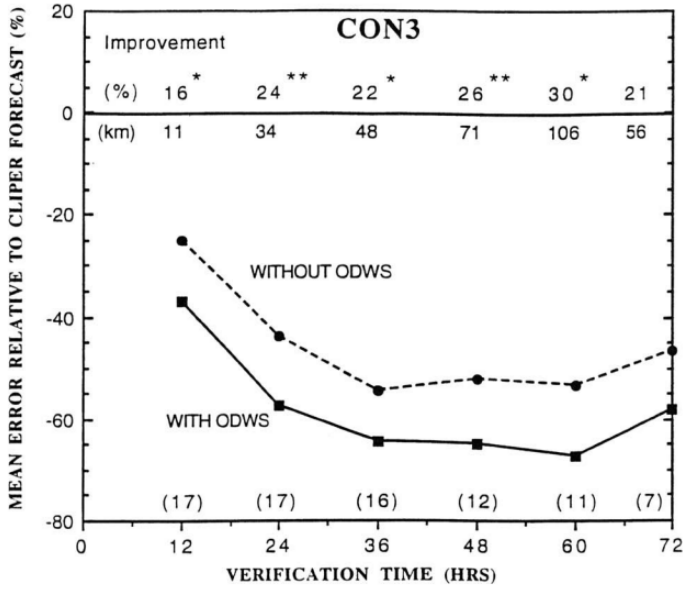
Data Denial Experiments: Dropsondes

- Burpee et al. (1996)
  - Utilized data from 28 Atlantic hurricane flights spanning 1982-1993, assimilating a total of 686 dropsondes
  - Evaluated using *three* leading models (VICBAR, GSM, GFDL). Results averaged together to form a consensus (CON3).
  - Hurricane track errors reduced by 16-30% in the 12-60-hour lead time
  - **A "breakthrough" in hurricane forecasting**

The Impact of Omega Dropwindsondes on Operational Hurricane Track Forecast Models



Robert W. Burpee,*,@ James L. Franklin,* Stephen J. Lord,+
Robert E. Tuleya,# and Sim D. Aberson*



Data Denial Experiments: Dropsondes

An Assessment of Omega Dropwindsonde Data in Track Forecasts of Hurricane Debby (1982)
Burpee et al., *1984*

The impact of omega dropwindsonde observations on barotropic hurricane track forecasts
Franklin and DeMaria, *1992*

Impact on Hurricane Track and Intensity Forecasts of GPS Dropwindsonde Observations from the First-Season Flights of the NOAA Gulfstream-IV Jet Aircraft
Aberson and Franklin, *1999*

Two Years of Operational Hurricane Synoptic Surveillance
Aberson, *2002*

Targeted observations to improve operational tropical cyclone track forecast guidance
Aberson, *2003*

Large Forecast Degradations due to Synoptic Surveillance during the 2004 and 2005 Hurricane Seasons
Aberson, *2008*

10 years of hurricane synoptic surveillance (1997–2006)
Aberson, *2010*

The Impact of Dropwindsonde Observations on Typhoon Track Forecasts in DOTSTAR and T-PARC
Chou et al., *2011*

The Impact of Dropwindsonde Data from the THORPEX Pacific Area Regional Campaign and the NOAA Hurricane Field Program on Tropical Cyclone Forecasts in the GFS
Aberson, *2011*

Assimilation of High-Resolution Tropical Cyclone Observations with an Ensemble Kalman Filter Using NOAA/AOML/HRD's HEDAS: Evaluation of the 2008–11 Vortex-Scale Analyses
Aksoy et al., *2013*

The Impact of Dropwindsonde and Supplemental Rawinsonde Observations on Track Forecasts for Hurricane Irene (2011)
Majumdar et al., *2013*

A Review of Targeted Observations
Majumdar, *2016*

Composite Impact of Global Hawk Unmanned Aircraft Dropwindsondes on Tropical Cyclone Analyses and Forecasts
Christopherson et al., *2018*

The Impact of NASA Global Hawk Unmanned Aircraft Dropwindsonde Observations on Tropical Cyclone Track, Intensity, and Structure: Case Studies
Christopherson et al., *2017*

Impact of UAS Global Hawk Dropsonde Data on Tropical and Extratropical Cyclone Forecasts in 2016
Kren et al., *2018*

Impact of Assimilating Aircraft Reconnaissance Observations on Tropical Cyclone Initialization and Prediction Using Operational HWRF and GSI Ensemble–Variational Hybrid Data Assimilation
Tong et al., *2018*

Impacts of Assimilating Additional Reconnaissance Data on Operational GFS Tropical Cyclone Forecasts
Sippel et al., *2022*

A Systematic Assessment of the Overall Dropsonde Impact during the 2017–20 Hurricane Seasons Using the Basin-Scale HWRF
Ditchek et al., *2023*

Data Denial Experiments: User Beware

- Although the addition of aircraft data, including dropsondes, generally leads to improved forecast model accuracy, it CAN lead to decreased model accuracy
- "Because of uncertainties in the data, the model, and the data assimilation, **some surveillance missions will necessarily lead to forecast degradations**."

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Large Forecast Degradations due to Synoptic Surveillance during the 2004 and 2005 Hurricane Seasons

SIM D. ABERSON

NOAA/AOML/Hurricane Research Division, Miami, Florida

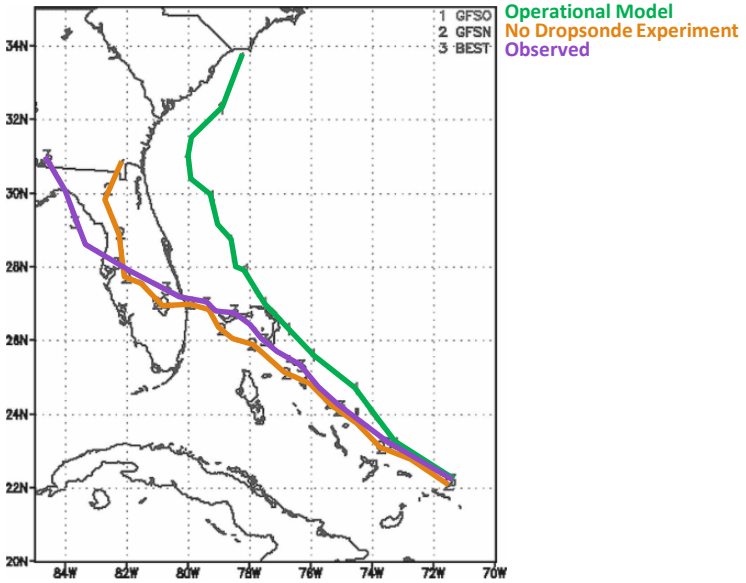


FIG. 1. GFS track forecasts for Hurricane Frances initialized at 0000 UTC 2 Sep 2004. Positions are plotted every 6 h through 5 days and are marked by a number every 12 h. BEST is the best track.

Data Denial Experiments: This Project

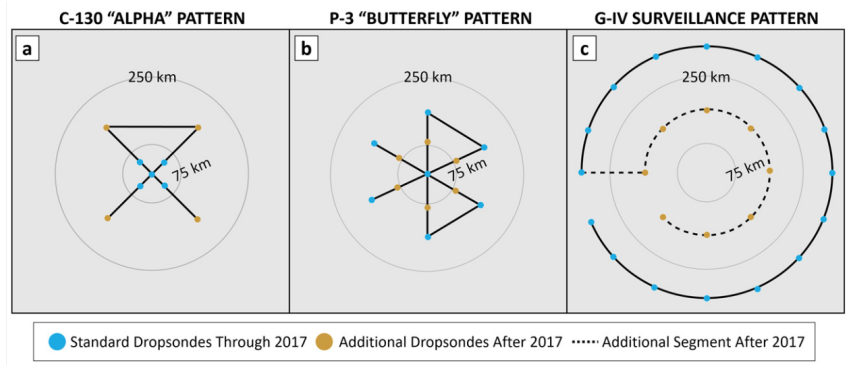
- For this project, we will utilize cases from **Ditchek et al. (2023)**
- Examined 92 tropical cyclones across Atlantic and East Pacific from 2017-2020
- 41 of the 92 TCs were sampled by dropsondes, and 27 of those were in the Atlantic
- Utilized the modern research-grade basin-scale HWRF model
- "**ALL**": all available data assimilated
- "**NO**": same as ALL, but dropsonde data omitted

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A Systematic Assessment of the Overall Dropsonde Impact during the 2017–20 Hurricane Seasons Using the Basin-Scale HWRF

SARAH D. DITCHEK,[a,b] JASON A. SIPPEL,[b] GHASSAN J. ALAKA JR.,[b] STANLEY B. GOLDENBERG,[b] AND LIDIA CUCURULL[b]

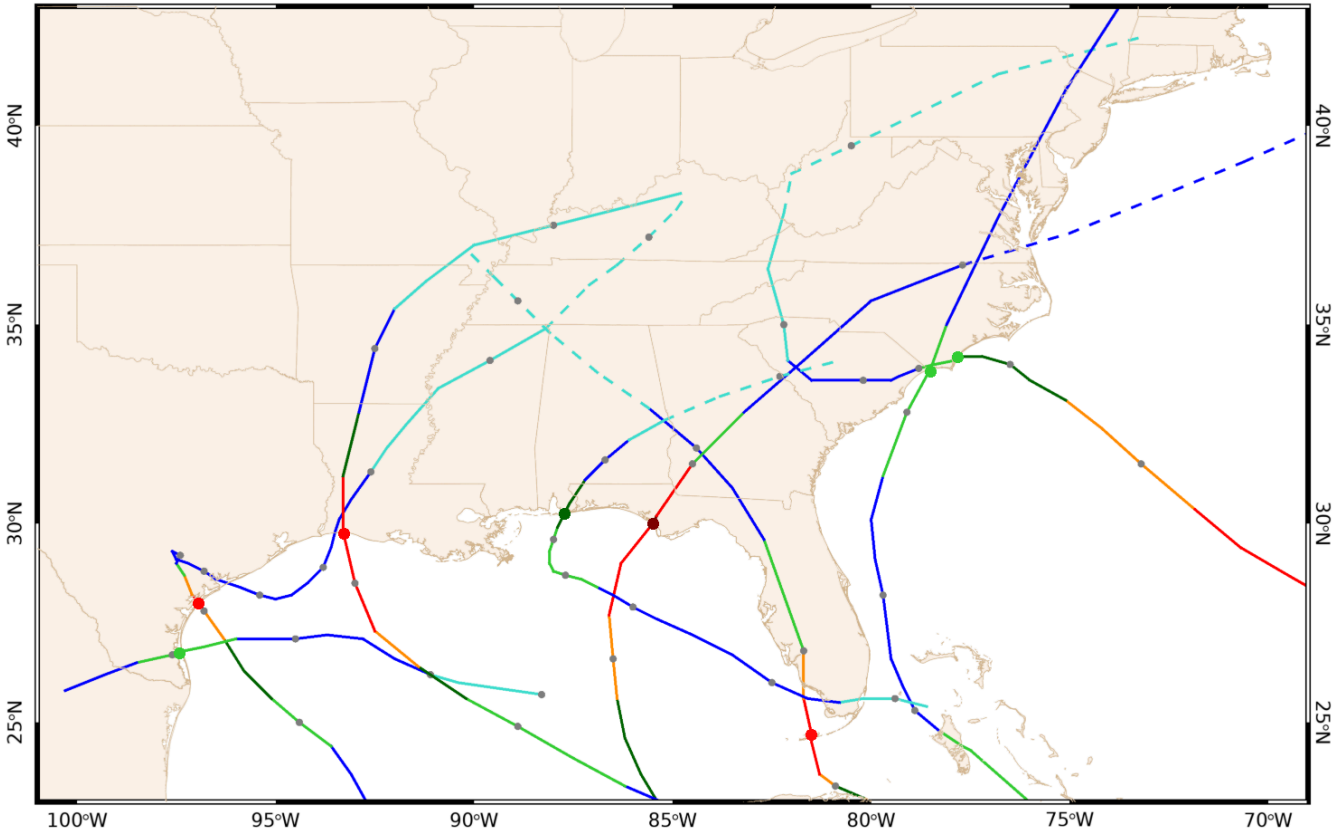
[a] *Cooperative Institute for Marine and Atmospheric Studies, University of Miami, Miami, Florida*
[b] *NOAA/AOML/Hurricane Research Division, Miami, Florida*



Data Denial Experiments: This Project

- From the 92 TCs in the Ditchek et al. 2023 comprehensive study, we select the 8 cases that made landfall in the U.S. at hurricane intensity.
  - Landfall intensities span Category 1-5, with half of the cases being classified as “major” (Category 3+).

Year	Storm Name	Landfall Intensity	Landfall Time
2017	Harvey	Category 4	26 Aug 0000
2017	Irma	Category 4	10 Sep 1200
2018	Florence	Category 1	14 Sep 0600
2018	Michael	Category 5	10 Oct 1200
2020	Hanna	Category 1	25 Jul 1800
2020	Isaias	Category 1	4 Aug 0000
2020	Laura	Category 4	27 Aug 0600
2020	Sally	Category 2	16 Sep 0600



Wind and Precipitation Modeling

- Produce 3 sets of wind and rainfall swaths for each model run
  - Select 5-day forecasts that are initialized 1, 2, and 3 days before landfall
  - Wind: "*stormwindmodel*" package (Andersen et al. 2018)
  - Rainfall: "*PHRaM*" (Lonfat et al. 2007), "*P-Rain*" (Marks et al. 2020)
  - ~ 84,000 total swaths created for our 8 selected cases
- Inputs to those models are based on:
  - NHC final best-track
  - ALL and NO model experiment outputs from Ditchek et al. (2023)
  - Vertical wind shear parameters from NHC's SHIPS model diagnostic files
  - 1000 ensemble members per forecast generated by NHC's wind speed probability model
    - Can then calculate ensemble mean and standard deviation of wind and rain
- Calculate county area-averages of all parameters

Examples -- From 1 day before Laura (2020) landfall

